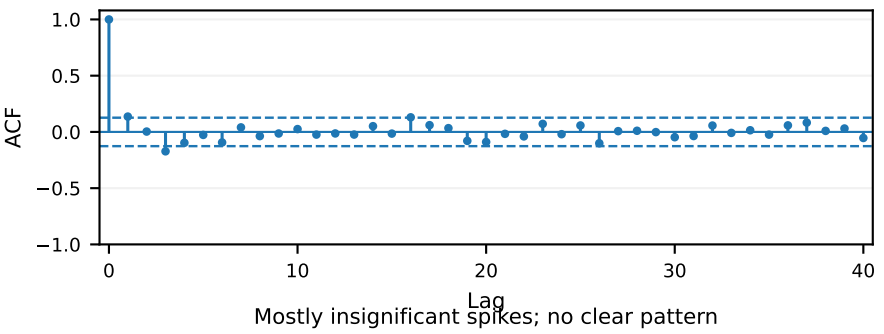
ACF Plot Comparison Guide

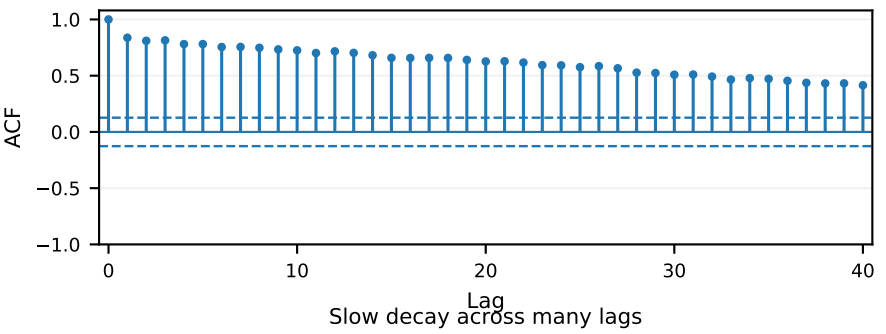
Visual patterns for matching ACF plots to time-series descriptions

A. White noise



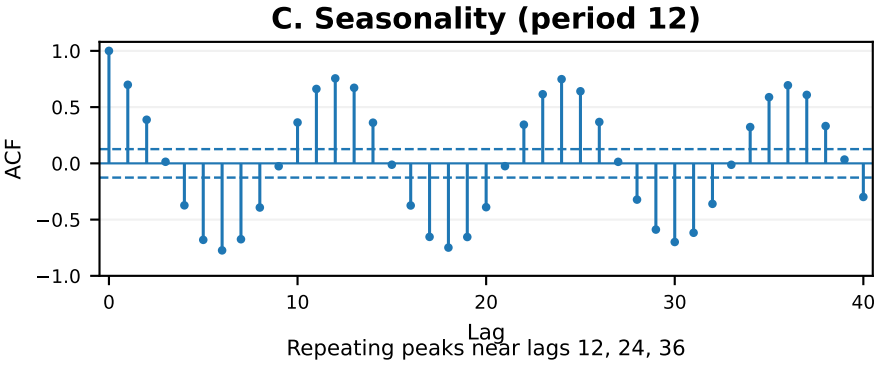
Mostly insignificant spikes; no clear pattern

B. Trend + noise



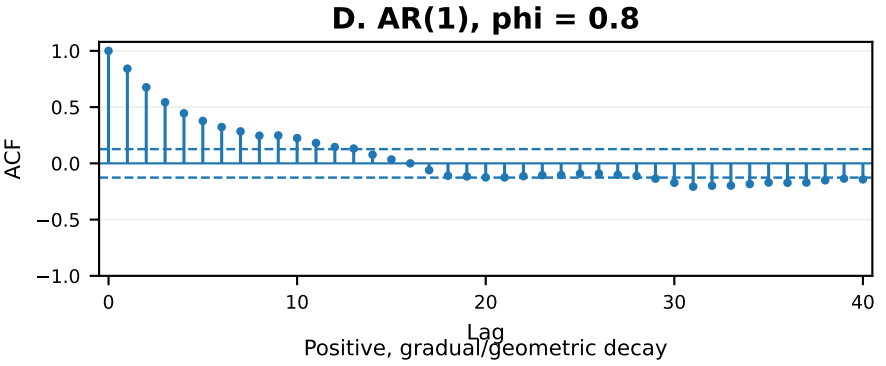
Slow decay across many lags

C. Seasonality (period 12)



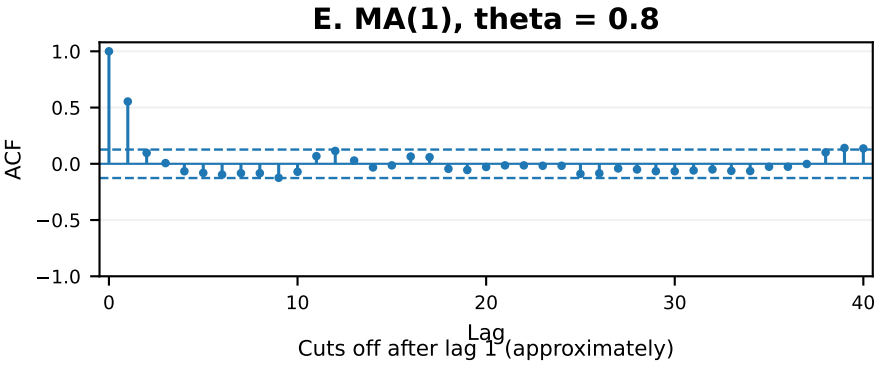
Repeating peaks near lags 12, 24, 36

D. AR(1), phi = 0.8



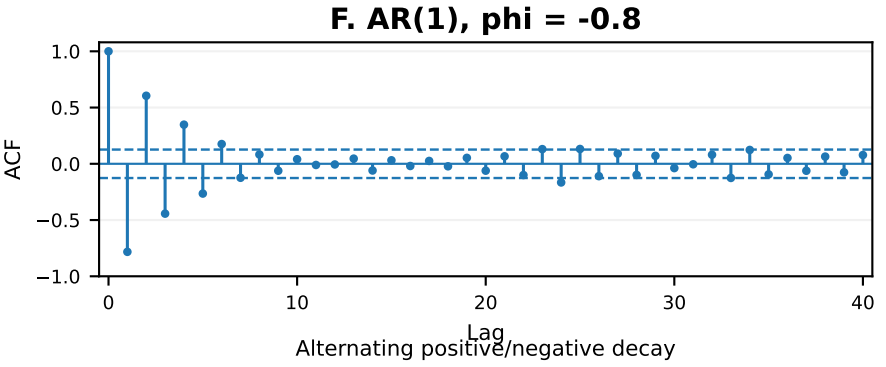
Positive, gradual/geometric decay

E. MA(1), theta = 0.8



Cuts off after lag 1 (approximately)

F. AR(1), phi = -0.8



Alternating positive/negative decay

ACF Shape Cheat Sheet

ACF pattern	What to think
Slow decay	Trend or non-stationarity
Exponential / gradual decay	AR-type process
Cuts off after lag q	MA(q) process
Alternating positive and negative	Oscillating behaviour or negative dependence
Spikes at seasonal lags	Seasonality (e.g. 12, 24, 36 for monthly data)
Most spikes inside significance bounds	White noise / little linear autocorrelation
Gradual decay with repeating peaks	Seasonality, possibly combined with trend

Dashed lines are approximate 95% bounds (±1.96/√n). Patterns are guides, not absolute rules.